Software Design Analysis Report

A. Project Details

Project title: Municipal Complaint Management System (MCMS)

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Project objective: To digitize civic grievance redressal by enabling citizens to report issues (sanitation, roads, streetlights, water supply) with photos and geolocation, allowing automated department routing, real-time ticket tracking, and SLA escalation. Technology Stack: Python, Flask/FastAPI, React, PostgreSQL/Supabase.

B. Module Identification and Modularization

Total number of modules: **4 Core Modules**. Explain why the project is divided this way. Include all major modules; add rows as required: The project is structured modularly by separating actor permissions (Citizen, Officer, Field Worker) and core data lifecycles (Authentication, Ingestion, Routing, Resolution) to enforce clean separation of concerns and maintainability.

Module ID	Module name	Main responsibility	Inputs / outputs	Depends on
M1	User & Role Management	Handles secure registration, login, and role verification.	In: Credentials Out: Auth Token	Supabase Auth
M2	Complaint Ingestion	Grievance submission, unique reference ID, GPS & media uploads.	In: Text, Image, GPS Out: Tracking ID	M1, Database
M3	Admin Routing & SLA	Routes complaints to departments and monitors SLA deadlines.	In: Open Tickets Out: Assigned Logs	M2 Data
M4	Field Resolution & Feedback	Field task status updates, star ratings, and citizen feedback.	In: Resolution Proof Out: Closed Status	M3 Workflow

Module hierarchy / decomposition diagram: MCMS Root ➔ [M1: Auth] | [M2: Ingestion] | [M3: Routing] | [M4: Field Ops]

C. Design Strategy

[ ] Top-Down [ ] Bottom-Up **[x] Hybrid**

Justification: Why is this strategy suitable for your project? A hybrid strategy is ideal because the overall municipal platform is structured top-down from high-level portals (Citizen, Officer, Admin) down to subsystems, while foundational components like database connection clients (`supabase_client.py`), JWT security verification, and media handlers are built bottom-up and integrated seamlessly.

Evidence: Decomposed from top portal layers down to service modules, backed by reusable low-level database utility wrappers.

D. Design Approach

[ ] Function-Oriented [ ] Object-Oriented **[x] Mixed**

Justification: Identify the main functions OR classes/objects and explain why this approach fits. The system uses a mixed approach combining Object-Oriented Programming (OOP) for domain entity modeling (`User`, `Citizen`, `Complaint`, and `Feedback` classes with structured attributes) alongside Function-Oriented RESTful controllers for processing CRUD API endpoints and state transitions efficiently.

Evidence: Entity classes manage internal data states while route handler functions manage incoming HTTP requests and responses.

E. Coupling Analysis

For at least three important module-to-module interactions, identify the information exchanged, dependency and coupling type (where applicable). Aim for low coupling.

Module pair	Interaction / shared data	Coupling type / level	How to reduce dependency
M2 ↔ M1	Passes verified `userId` to bind grievances to citizen profiles.	Data Coupling (Low)	Pass minimal JWT payload.
M3 ↔ M2	Reads and updates complaint status flags and department codes.	Stamp Coupling (Medium)	Encapsulate state logic.
M4 ↔ M3	Updates worker assignment logs and completion timestamps.	Control Coupling (Low)	Use event-driven flags.

F. Cohesion Analysis

For at least three modules, describe the single main responsibility and evaluate how closely its internal tasks belong together. Aim for high cohesion.

Module	Main responsibility	Cohesion type / level	Improvement
M1: Auth Module	Managing user login, session tokens, and role validation.	Functional Cohesion (High)	Isolate checks.
M2: Complaint Module	Grievance form processing, photo storage, and geolocation.	Sequential Cohesion (High)	Isolate media upload.
M4: Feedback Module	Capturing field worker proof, star ratings, and feedback.	Functional Cohesion (High)	None needed.

G. Module Interaction and Design Quality

Module Interaction Diagram / Sequence Overview: Citizen logs in via M1, submits geo-tagged grievance with photos through M2, receiving a unique tracking ID. M3 routes the ticket to the municipal department where an officer assigns it to ground staff. Field Worker resolves and updates via M4, prompting the Citizen to submit star ratings and final feedback upon completion.